

1) Consider the CMOS inverter with the following parameters:

$$nmos: V_{Tn} = 0.6V \quad \mu_n C_{ox} = 60 \mu A/V^2 \quad (W/L)_n = 8$$

$$pmos: V_{Tp} = -0.7 \quad \mu_p C_{ox} = 25 \mu A/V^2 \quad (W/L)_p = 12$$

Calculate the noise margin and the switching threshold (V_{th}) of this circuit, the power supply voltage $V_{DD} = 3.3V$!

- switching threshold (output = Input)
- NMOS & PMOS are ON (saturation region)
- Drain current are equal $I_{Dn} = I_{Dp}$

We know that

$$I_{Dn} = \frac{1}{2} \mu_n C_{ox} \left(\frac{W}{L} \right)_n (V_{Gn} - V_{Tn})^2$$

$$= \frac{1}{2} (60)(8) (V_{Gn} - 0.6)^2$$

$$I_{Dn} = 240 (V_{Gn} - 0.6)^2 \rightarrow (1)$$

$$I_{Dp} = \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right)_p (V_{Gp} - V_{Tp})^2$$

$$= \frac{1}{2} \mu_p C_{ox} \left(\frac{W}{L} \right)_p (V_{DD} - V_{in} - V_{Tp})^2$$

$$= \frac{1}{2} (25)(12) (3.3 - V_{in} - 0.7)^2$$

$$= 150 (2.6 - V_{in})^2 \rightarrow (2)$$

$$\therefore [V_{in} = V_{out}]$$

Equate (1) = (2)

$$240 (V_{Gn} - 0.6)^2 = 150 (2.6 - V_{Gn})^2$$

$$\sqrt{8} (V_{Gn} - 0.6)^2 = \sqrt{5} (2.6 - V_{Gn})$$

$$\boxed{V_{Gn} = 1.248V}$$

Noise Margin $\rightarrow$

$$V_{OH} \simeq V_{DD} = 3.3V$$

$$V_{OL} = 0V$$

Threshold volt for noise margin

$$V_{IL} \simeq V_{TN} = 0.6V$$

$$V_{IH} \simeq V_{DD} - |V_{TP}| = 3.3 - 0.7 = 2.6$$

1) Noise margin low (NML):

$$NML = V_{IH} - V_{OL}$$

$$= 0.6 - 0$$

$$\boxed{NML = 0.6}$$

2) NM_H

$$NM_H = V_{OH} - V_{IH}$$

$$= 3.3 - 2.6$$

$$\boxed{NM_H = 0.7}$$